

Mila Ardath

Squamish, BC, Canada · amelia.ardath@gmail.com · github.com/CtrlSpice

I'm a software engineer who builds developer tools. The current one is `otel-desktop-viewer`, which shows you your OpenTelemetry data in a browser without standing up a backend. I've worked the whole stack, embedded to observability. I like building genuinely useful things, and the problems never stop being interesting.

Skills

Languages: Go, TypeScript, JavaScript, SQL

Telemetry: OpenTelemetry, OTLP, distributed tracing, observability

APIs: gRPC, REST, JSON-RPC

Data: DuckDB, PostgreSQL, analytical schema design

Frontend & tooling: Svelte, React, CodeMirror/Lezer query language design, CI/CD

Experience

Open Source Maintainer, `otel-desktop-viewer` — 2023–present

github.com/CtrlSpice/otel-desktop-viewer

1,210 stars, 53 forks, and 4,900+ release binaries downloaded from GitHub, plus installs through Homebrew, Docker, and `go install`.

- Architected it as a custom OpenTelemetry Collector distribution: an exporter that writes telemetry, and a collector extension that owns the DuckDB store and the web UI, so storage outlives the pipeline
- Designed the analytical schema for traces, metrics, and logs, including a content-addressed attribute dictionary that dedupes attributes at ingest
- Implemented metric reduction and OTLP exponential-histogram merging in SQL, where a wrong answer still renders as a plausible chart — found four such bugs, and now verify the results against an independent implementation over randomized inputs
- Rewrote the frontend in Svelte 5 and TypeScript, replacing the original React implementation, and built a search language with a Lezer grammar in CodeMirror that the Go backend compiles into SQL
- Built the CI/CD release pipeline: a GitHub Actions matrix on native runners per platform (CGO rules out cross-compilation), publishing to Homebrew, GHCR, deb/rpm, and GitHub Releases across macOS, Linux, and Windows

Senior Software Engineer, Telus — 2023–2025

Internal platform organization; deployment automation for services running on Kubernetes.

- Designed and implemented the tick-based state machine at the core of a tool automating blue-green and canary deployments
- Built a golden-signal metrics comparator used as a deploy gate, so only artifacts with stable metrics promote to production
- Drove OpenTelemetry adoption across teams: onboarded engineers new to OTel, paired with them on instrumenting their services, and closed instrumentation gaps in shared infrastructure — usually trace headers stripped in transit
- Technologies: Go, Kubernetes, client-go, PostgreSQL

Career break — 2016–2022

Firmware Testing and Integration Developer, Ciena — 2014–2016

- Wrote white- and black-box test suites for firmware on fiber-optic network equipment, built network simulation tooling to reproduce configuration errors, and thermal-cycled switches in environmental chambers to see how they'd do in a desert
- Technologies: C, C++, TCL, Perl, Bash

Developer, Applied Research and Innovation, Algonquin College — 2012–2014

- Prototyped pro-athlete vision-training drills as a Unity game with an optometrist client, to see how they could be made accessible to more people; designed a backend-as-a-service for rapid prototyping
- Technologies: C#, Unity, Java, Android, MySQL

Education

- Recurse Center (remote) — 2025–2026
Self-directed programming retreat, admitted by application. Completed 18 weeks across two batches.
- Engineering Technology and Computer Science, Algonquin College — 2014